

Assigning Static IPs to NLBs with ALB Target Groups

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Introduction

In this lab, I configured a multi-tier Elastic Load Balancer (ELB) architecture to combine the intelligent routing capabilities of an Application Load Balancer (ALB) with the static IP addressing capabilities of a Network Load Balancer (NLB). This approach allows external DNS systems, which cannot use AWS-specific alias records like those available in Route 53, to reference load balancers through consistent IPs.

1. Created a Target Group for the ALB

The screenshot displays the AWS Management Console for the EC2 service in the United States (N. Virginia) Region. A yellow banner at the top indicates a local storage issue. The left sidebar shows navigation options under 'EC2', including 'Instances', 'Images', 'Elastic Block Store', and 'Network & Security'. The main content area is divided into several sections:

- Resources:** A table showing counts for various EC2 resources.

Resource	Count
Instances (running)	3
Dedicated Hosts	0
Key pairs	0
Security groups	4
Auto Scaling Groups	1
Elastic IPs	1
Load balancers	1
Snapshots	0
Capacity Reservations	0
Instances	3
Placement groups	0
Volumes	3
- Launch instance:** A section with a 'Launch instance' button and a 'Migrate a server' link. It includes a note about the region.
- Instance alarms:** A section showing '0 in alarm', '0 OK', and '0 insufficient data'.
- Service health:** A section showing the region as 'United States (N. Virginia)' and a status of 'This service is operating normally.' It also lists available zones.
- Account attributes:** A section showing default VPC, settings, and account information.
- Explore AWS:** A section with promotional links for Spot Instances, Graviton2, and cost reduction.

Assuming an existing Auto Scaling Group (ASG) was already running with two EC2 instances behind an ALB, I proceeded to create a target group pointing to this ALB.

The screenshot shows the 'Specify group details' page in the AWS Management Console. The page is divided into several sections:

- Step 1: Specify group details** (selected) and **Step 2: Register targets** (disabled).
- Basic configuration**: A section where settings cannot be changed after creation. It includes a 'Choose a target type' section with three options:
 - Instances**: Supports load balancing to instances within a specific VPC and facilitates the use of Amazon EC2 Auto Scaling.
 - IP addresses**: Supports load balancing to VPC and on-premises resources, facilitates routing to multiple IP addresses, and offers flexibility with microservice-based architectures.
 - Lambda function**: Facilitates routing to a single Lambda function and is accessible to Application Load Balancers only.
 - Application Load Balancer** (selected): Offers flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC and facilitates using static IP addresses and PrivateLink with an Application Load Balancer.
- Target group name**: A text input field containing 'ApplicationLoadBalancer-tg'. A note states: 'A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.'
- Protocol: Port**: A section where a protocol is chosen for the target group. The protocol is set to 'TCP' and the port is '80'. A note states: 'Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation.'
- VPC**: A section where the VPC is selected. The selected VPC is 'Your Custom VPC' with ID 'vpc-0d7199f75ee0236e' and IPv4 CIDR '10.0.0.0/16'. A note states: 'Select the VPC with the Application Load Balancer that you want to include in the target group.'
- Health checks**: A section where the associated load balancer periodically sends requests to the registered targets. It includes a 'Health check protocol' dropdown set to 'HTTP' and a 'Health check path' text input field containing '/'. A note states: 'Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred. Up to 1024 characters allowed.'
- Advanced health check settings**: A link to expand the health check settings.
- Attributes**: A section where certain default attributes will be applied to the target group. A note states: 'Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.'
- Tags - optional**: A section where tags can be added to the target group. A note states: 'Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.'

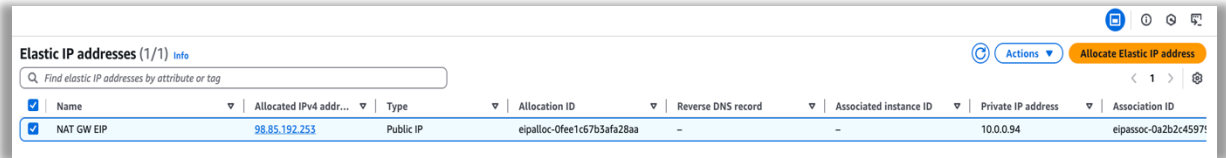
At the bottom right, there are 'Cancel' and 'Next' buttons.

- In the EC2 console, under Load Balancing > Target Groups, I selected Create target group.
- For the target type, I chose Application Load Balancer.
- I named the target group ApplicationLoadBalancer-tg.
- Set the Protocol to TCP and Port to 80.
- Selected the appropriate custom VPC from the dropdown list.
- For health checks, I configured the protocol as HTTP and set the path to /.
- Proceeded to register the target by selecting Register now.
- From the Application Load Balancer dropdown, I selected OurApplicationLoadBalancer.
- Clicked Create target group to complete the process.

2. Allocated Elastic IP Addresses (EIPs)

To ensure static IP addressing for the NLB, I allocated Elastic IPs:

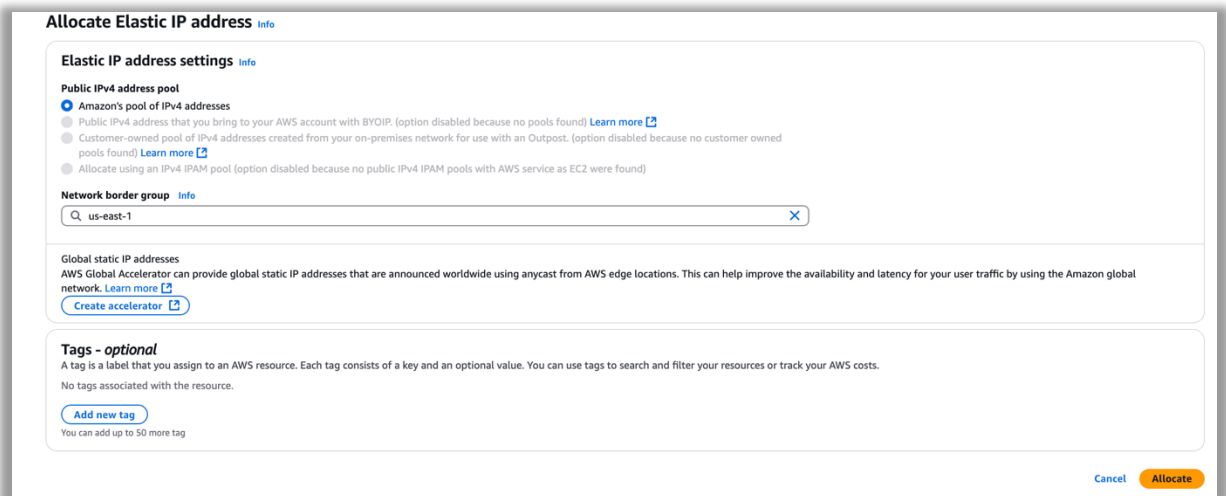
Navigated to Elastic IPs under Network & Security.



The screenshot shows the 'Elastic IP addresses' page in the AWS console. It displays a table with one entry: a NAT GW EIP with an allocated IPv4 address of 98.85.192.253. The table columns include Name, Allocated IPv4 address, Type, Allocation ID, Reverse DNS record, Associated instance ID, Private IP address, and Association ID.

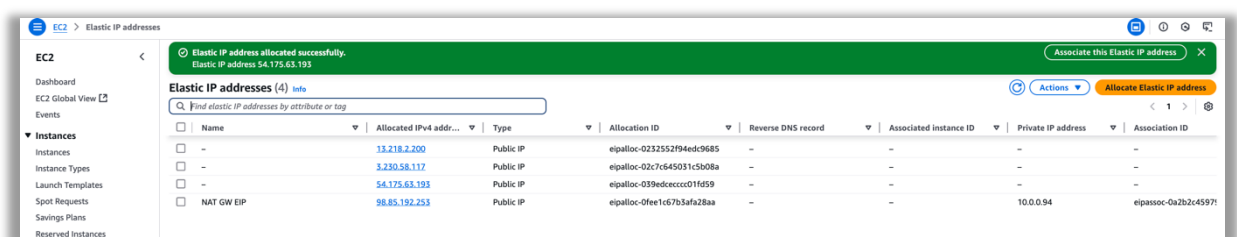
Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated instance ID	Private IP address	Association ID
NAT GW EIP	98.85.192.253	Public IP	eipalloc-0fee1c67b3afa28aa	-	-	10.0.0.94	eipassoc-0a2b2c4597f

Clicked Allocate Elastic IP address.



The screenshot shows the 'Allocate Elastic IP address' dialog box. It includes sections for 'Elastic IP address settings', 'Network border group' (set to us-east-1), 'Global static IP addresses', and 'Tags - optional'. The 'Public IPv4 address pool' section shows 'Amazon's pool of IPv4 addresses' selected. The 'Allocate' button is highlighted in orange.

- Ensured the Network Border Group was set to us-east-1.
- Left Amazon's pool of IPv4 addresses selected.
- Repeated the allocation process three times, recording each EIP for later use.



The screenshot shows the 'Elastic IP addresses' page with a green notification banner stating 'Elastic IP address allocated successfully. Elastic IP address 54.175.63.193'. The table now shows four entries: three public IP addresses and one NAT GW EIP.

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated instance ID	Private IP address	Association ID
-	13.218.2.200	Public IP	eipalloc-0232552f94ed9685	-	-	-	-
-	3.230.58.117	Public IP	eipalloc-02c7c645031c5b08a	-	-	-	-
-	54.175.63.193	Public IP	eipalloc-039edceccc01f659	-	-	-	-
NAT GW EIP	98.85.192.253	Public IP	eipalloc-0fee1c67b3afa28aa	-	-	10.0.0.94	eipassoc-0a2b2c4597f

3. Created the Network Load Balancer (NLB) and Listener

With EIPs and a target group ready, I proceeded to configure the NLB:

- In the Load Balancers section of the EC2 console, I clicked Create load balancer and chose Network Load Balancer.
- Named it WebServer-nlb.
- Selected Internet-facing for the scheme.
- Kept IPv4 for IP address type.

The screenshot shows the 'Create Network Load Balancer' page in the AWS Management Console. The breadcrumb navigation at the top reads 'EC2 > Load balancers > Create Network Load Balancer'. A blue notification banner at the top states: 'Network Load Balancer now supports UDP for Dualstack. Set your IP address type as dualstack and enable prefix for IPv6 source NAT. Then configure UDP-based listeners to route to IPv6 targets.' Below this is a section titled 'How Network Load Balancers work'. The main configuration area is titled 'Basic configuration' and includes three sections: 'Load balancer name' with a text input field containing 'WebServer-nlb'; 'Scheme' with two radio button options, 'Internet-facing' (selected) and 'Internal'; and 'Load balancer IP address type' with two radio button options, 'IPv4' (selected) and 'Dualstack'. At the bottom, the 'Network mapping' section shows a dropdown menu for 'Your Custom VPC' with the value 'vpc-047199f75ee0236a' selected. The page includes various informational links and a close button for the notification banner.

Network Mapping

- Selected the custom VPC (10.0.0.0/16).
- Enabled all three Availability Zones and selected a Public Subnet for each.
- For each subnet, selected Use an Elastic IP address, then assigned a unique EIP from the ones allocated earlier.

Security Groups and Listener

- In the Security groups section, selected the AllowTCPFromEverywhere group and removed the default.
- Under Listeners and Routing, kept the listener configuration at TCP:80.
- For the default action, selected the previously created ApplicationLoadBalancer-tg.
- Clicked Create load balancer and waited for WebServer-nlb to reach the Active state.

Network mapping [info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to on-premises targets or if using VPC peering. To confirm the VPC for your targets, view [target groups](#). For a new VPC, [create a VPC](#).

Your Custom VPC

vpc-0d7199f75ee0236e

IPv4 VPC CIDR: 10.0.0.0/16

Availability Zones and subnets

Select one or more Availability Zones and corresponding subnets. Enabling multiple Availability Zones increases the fault tolerance of your applications. The load balancer routes traffic to targets in the selected Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

us-east-1a (use1-az6)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0c9523f6c10fb2abf

IPv4 subnet CIDR: 10.0.0.0/24

Public Subnet AZ A

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

Assigned by AWS

Use an Elastic IP address

IP address

Specify an elastic IP address to provide your load balancer with a static IPv4 address in the selected Availability Zone.

13.218.2.200

elpalloc-0232552f94edc9685

us-east-1b (use1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-095e561d604cdf352

IPv4 subnet CIDR: 10.0.1.0/24

Public Subnet AZ B

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

Assigned by AWS

Use an Elastic IP address

IP address

Specify an elastic IP address to provide your load balancer with a static IPv4 address in the selected Availability Zone.

3.230.58.117

elpalloc-02c7c645031c5d08a

us-east-1c (use1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0d5efac9401256d40

IPv4 subnet CIDR: 10.0.2.0/24

Public Subnet AZ C

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

Assigned by AWS

Use an Elastic IP address

IP address

Specify an elastic IP address to provide your load balancer with a static IPv4 address in the selected Availability Zone.

54.175.63.193

elpalloc-039edc0cc01fd59

Security groups [info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups - recommended

Security groups support on Network Load Balancers can only be enabled at creation by including at least one security group. You can change security groups after creation. The security groups for your load balancer must allow it to communicate with registered targets on both the listener port and the health check port. For PrivateLink Network Load Balancers, security group rules are enforced on PrivateLink traffic; however, you can turn off inbound rule evaluation after creation within the load balancer's Security tab or using the API.

Select up to 5 security groups

AllowTCPFromEverywhere

sg-04f7f101a30b3f6800

VPC: vpc-0d7199f75ee0236e

Listeners and routing [info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener TCP:80

Remove

Protocol

TCP

Port

80

1-65535

Default action [info](#)

Forward to

ApplicationLoadBalancer-tg

TCP

Target type: Application Load Balancer, IPv4

Create target group

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Add listener

4. Verified Functionality

Once the NLB became active:

- I copied the DNS name of WebServer-nlb and accessed it via a browser to confirm proper functionality.
- Navigated back to the EC2 > Load Balancers section, selected the NLB, and under the Listeners tab, opened the associated Target Group in a new tab.

Load balancers (2)							Actions	Create load balancer
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.								
Filter load balancers							< 1 >	
☐	Name	DNS name	State	VPC ID	Availability Zones	Type	Date created	
☐	WebServer-nlb	WebServer-nlb-134d26382...	Active	vpc-0d7199f75eee0236e	3 Availability Zones	network	May 22, 2025, 20:15 (UTC+01:00)	
☐	OurApplicationLoadBal...	OurApplicationLoadBalance...	Active	vpc-0d7199f75eee0236e	3 Availability Zones	application	May 22, 2025, 19:42 (UTC+01:00)	

Verified that the target health status was marked as Healthy, confirming that the NLB was successfully routing traffic to the ALB and onward to the EC2 instances behind it.

ApplicationLoadBalancer-tg				Actions
Details arn:aws:elasticloadbalancing:us-east-1:459559510335:targetgroup/ApplicationLoadBalancer-tg/744ca722badf4b3b Target type Application Load Balancer Protocol : Port TCP: 80 VPC vpc-0d7199f75eee0236e IP address type IPv4				
Load balancer WebServer-nlb WebServer-nlb is routing traffic to this Application Load Balancer target group. You can now use static IP addresses, enable AWS PrivateLink, and route multi-protocol connections.				
Targets Health checks Attributes Tags				
Registered target Application Load Balancer target groups are limited to a single Application Load Balancer target. The load balancer starts routing requests to a newly registered target as soon as the registration process is completed and the target passes initial health checks.				Deregister
Application Load Balancer OurApplicationLoadBalancer Port 80 ARN arn:aws:elasticloadbalancing:us-east-1:459559510335:loadbalancer/app/OurApplicationLoadBalancer/996c26b39...		Health status Healthy Health status details		

Conclusion

This setup successfully leveraged a Network Load Balancer to provide static IP addresses for external DNS systems, while routing traffic to an Application Load Balancer capable of intelligent HTTP routing. The architecture resolved the challenge of integrating with DNS platforms that cannot use AWS ALB alias records, enabling static, reliable endpoint management for web applications hosted on EC2 behind an ALB.

Hello Gurus!

I live in this Availability Zone: us-east-1b

I go by this Instance Id: i-0007bb3c5f6e4a395

Hello Gurus!

Back

I live in this Availability Zone: us-east-1a

I go by this Instance Id: i-0e85434d3c7dcfaee

Hello Gurus!

I live in this Availability Zone: us-east-1a

I go by this Instance Id: i-0e85434d3c7dcfaee